

JAYANTH GUNTUKU

Full-Stack Software Engineer (4+ YOE)

+91-9573419011 guntukujayanth35@gmail.com [LinkedIn: Jayanth Guntuku](https://www.linkedin.com/in/jayanthguntuku) [GitHub: Jayanth Guntuku](https://github.com/JayanthGuntuku) Texas, USA

WORK EXPERIENCE

Synapse ITS, USA | Software Engineer

Dec 2024 - Present

Tech Stack: React, Golang, JavaScript, REST APIs, WebSocket, MySQL, Docker, GCP, AWS

- Engineered and shipped 5+ full-stack features end-to-end using React on the frontend and Go on the backend, **improving page load performance by up to 40%**.
- Architected and maintained 30+ REST APIs in Go consumed by the React frontend, with backend logic covering end-to-end data processing and business rules, **reducing integration issues by 30%**.

Platform & Access Management

- Delivered a Go-based gateway service bridging cloud infrastructure and 1000+ edge devices across 100+ agencies, implementing bidirectional message routing and **enabling 100% remote access** where none previously existed.
- Rolled out subscription-based access control and RBAC across the platform, **reducing support tickets by 20%** and simplifying onboarding for 100+ agencies.

Syracuse University, USA | Software Intern + Graduate Teaching Assistant

Aug 2023 - Nov 2024

Tech Stack: Java, Spring Boot, React, MySQL, Google Maps API, Apache Kafka, Docker, Kubernetes, AWS

- Spearheaded a full-stack **real-time campus shuttle tracking** app using Java Spring Boot backend and React frontend, **integrating Google Maps API** to serve live locations and ETAs to 1000+ daily users.

System Design & Scalability

- Deployed a predictive model on historical and real-time data to optimize routes, **cutting average wait times by 30%** and **increasing shuttle usage by 20%**.
- Architected the system for scalability using Apache Kafka for event-driven data processing and WebSocket for real-time delivery, achieving **99% uptime during peak hours**.

Infosys, India | Systems Engineer

Aug 2020 - May 2022

Tech Stack: Java, Spring Boot, React, REST APIs, SQL Server, MySQL, JUnit, AWS

- Delivered and enhanced enterprise full-stack web applications using Java Spring Boot backend and React frontend, shipping scalable REST APIs and modern UI that **increased user engagement by 25%**.
- Engineered and maintained 50+ REST APIs in Java Spring Boot; achieved 95% test coverage using JUnit and Jasmine, **cutting production bugs by 30%** and **response times by 20%**.
- Optimized SQL Server and MySQL schemas with Hibernate ORM, cutting query time by 15-20%; deployed on AWS (EC2, RDS, S3, Lambda), **improving load times by 30%**.

PROJECTS

Image Super Resolution CNN

- Engineered a 6-layer deep CNN with residual learning in PyTorch, trained on 500 BSD images across 300 epochs on T4 GPU, achieving 30.60 dB PSNR and 0.8961 SSIM on 200 test images, surpassing the 30 dB industry threshold.
- Built a custom degradation pipeline generating 128x128 training pairs, improving PSNR by +1.67 dB and SSIM by +4.8% over baseline; replaced MSELoss with L1Loss and StepLR scheduling, reducing validation loss by 78%.

Online Examination Portal

- Delivered an end-to-end full-stack exam platform with React frontend, Java Spring Boot backend, RBAC, adaptive difficulty, and automated grading, reducing admin time by 50% and improving average grades by 10%.
- Integrated asynchronous processing via Apache Kafka and load balancing for scalability, cutting question disputes by 20% and boosting academic support usage by 20%.

EDUCATION

M.S. in Computer and Information Science | Syracuse University, USA | CGPA: 3.9/4.0

Aug 2022 - May 2024

B.Tech in Computer Science and Engineering | Gayatri Vidya Parishad College of Engineering, Vizag, India |

Aug 2016 - Aug 2020

CGPA: 8.43/10.0

TECHNOLOGIES & SKILLS

- Frontend:** React.js, TypeScript, JavaScript, HTML5, CSS3, Bootstrap, State Management
- Backend:** Node.js, Golang, .NET Core, REST APIs, Microservices, WebSocket
- Cloud & DevOps:** AWS (EC2, S3, Lambda, RDS), GCP, Docker, Kubernetes, CI/CD, Git
- AI/ML:** PyTorch, CNN, Deep Learning, Computer Vision, Image Processing, NumPy, scikit-image, GPU Computing (CUDA)
- Testing & Quality:** Jest, React Testing Library, NUnit, Jasmine, Unit Testing, Integration Testing
- Databases:** MySQL, SQL Server, MongoDB, DynamoDB
- Languages:** JavaScript, TypeScript, Golang, Python, Java, C++